# GAUGE-ALIGNED ENDPOINT INTERFACES FOR CROSS-DIMENSIONAL TEXT EMBEDDING DISTILLATION

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## ABSTRACT

Embedding distillation transfers knowledge from a large teacher to a smaller student, but their representations often differ in dimension and orientation. We propose GATE-KD, a final-embedding distillation method built around a comparable endpoint interface. A fixed principal-component projection selects a teacher subspace at the student dimension, while closed-form orthogonal Procrustes alignment follows the evolving student across epochs without altering the projected teacher geometry. We augment this interface with a native-space topological regularizer that matches degree-zero connectivity scales without requiring equal embedding dimensions. GATE-KD uses only cached final teacher embeddings and introduces no trainable alignment module. Under a label-free task-adapted protocol spanning three teacher–student configurations and nine sentence-level benchmarks, GATE-KD obtains the numerically highest aggregate score among the evaluated distilled students, exceeding the next-best baseline by 0.38–1.59 points while reducing measured GPU optimization-loop time by 62–74%. Controlled analyses identify coordinate alignment as the primary mechanism, while the topological term provides a smaller complementary regularization effect. These results support a gauge-aligned endpoint interface as a competitive alternative to more complex pipelines in the task-adapted sentence-level regime studied here.

## 1 INTRODUCTION

Text embedding models support retrieval, semantic similarity, clustering, and classification, but the strongest encoders are often too costly for latency- or memory-constrained deployment. Knowledge distillation offers a natural route to compact embedding models by transferring supervision from a frozen teacher to a smaller student (Hinton et al., 2015; Reimers & Gurevych, 2020). Unlike class logits, however, embedding coordinates have no shared semantics across models. Teacher and student representations may differ in dimension, orientation, depth, and even architecture. Existing methods address this mismatch with learned projectors, relational or contrastive objectives, and supervision across intermediate layers (Park et al., 2019; Tian et al., 2020; Xu et al., 2023; Zhang et al., 2024a; Truong et al., 2025; Dao et al., 2026). Intermediate alignment can provide rich supervision, but it presupposes useful correspondences between teacher and student depths—an assumption that becomes fragile as their architectures diverge—and often requires additional mappings or repeated access to teacher hidden states (Ko et al., 2023; Yu et al., 2025). More fundamentally, this machinery can obscure whether the difficulty lies in coordinating the representation hierarchy or in failing to define a comparable final representation in the first place. This motivates our hypothesis that the endpoint interface is an under-examined bottleneck in cross-dimensional distillation.

We test this hypothesis by centering the transfer mechanism on an *extrinsic interface* that expresses teacher representations in coordinates the student can match pointwise. We augment this primary mechanism with an *intrinsic regularizer* defined on properties of the representation cloud that do not depend on its coordinates or ambient dimension. This decomposition leads to **GATE-KD**, Gauge Alignment and Topology for Embedding Distillation. GATE-KD fits a fixed principal-component subspace to cached teacher endpoints, selecting the information that can pass through the student-width bottleneck. Within that subspace, an orthogonal Procrustes update aligns the target coordinates to the current student and is refitted after each epoch on a fixed calibration subset. Because each update is orthogonal, it changes only the targets' coordinate gauge, leaving their pairwise geometry unchanged. Because projection cannot preserve all teacher geometry, GATE-KD additionally uses a degree-zero persistent-homology ($H_0$) regularizer that matches distributions of batchwise connectivity scales in the original teacher and student spaces. This auxiliary term is coordinate-invariant and applies directly across different embedding dimensions. The resulting objective combines primary pointwise endpoint correspondence with coordinate-invariant structural regularization while supervising only the final student embedding. It trains no projector, uses no intermediate-layer objective, and requires no online teacher forward pass during ordinary training.

Experiments across three teacher–student configurations and nine sentence-level benchmarks under a label-free task-adapted protocol support this endpoint-centered view. Controlled decompositions show that the Procrustes-aligned gauge improves endpoint supervision beyond retaining teacher variance alone, while native-space $H_0$ regularization provides a complementary refinement. Overall, GATE-KD obtains the numerically highest aggregate score among the evaluated distilled students in all three configurations, exceeding TALAS by 0.38, 0.48, and 1.59 points while using only 26–38% of its measured GPU optimization time. These results show that endpoint-only supervision can be competitive with the evaluated hierarchy-based alternatives in the task-adapted sentence-level settings studied here.

We make three contributions.

- We formulate cross-dimensional embedding distillation as an intrinsic–extrinsic decomposition, separating pointwise coordinate comparability from coordinate-invariant structural regularization.
- We introduce GATE-KD, combining fixed PCA, epoch-wise closed-form Procrustes alignment, and native-space $H_0$ regularization using only cached teacher endpoints.
- Across three teacher–student configurations, GATE-KD obtains the highest numerical aggregate among the evaluated baselines while reducing measured optimization-loop cost; controlled analyses identify coordinate comparability as the primary source of improvement.

## 2 Related Work

Early text-embedding distillation directly regressed student sentence vectors onto teacher outputs, showing that unlabeled text can provide dense supervision without task annotations (Reimers & Gurevych, 2020). Later methods enrich this signal through contrastive objectives, specialized training recipes, and teacher-compatible output spaces (Gao et al., 2021; Xu et al., 2023; Xie et al., 2023; Zhang et al., 2024a; Li et al., 2025; Vujanic & Rückstieß, 2026). DistilCSE transfers a contrastive teacher through unlabeled-data distillation, whereas SimTDE combines token- and sentence-level objectives with a projection layer that relaxes equal-width requirements. LEALLA studies lightweight, low-dimensional sentence encoders in a multilingual regime (Mao & Nakagawa, 2023). Data-centric systems such as Gecko and jina-embeddings-v5-text additionally use generated, relabeled, or task-targeted training data (Lee et al., 2024; Akram et al., 2026); GATE-KD instead holds the unlabeled corpus and downstream protocol fixed to isolate the teacher–student interface. EmbedDistill combines endpoint matching with relative geometric supervision for retrieval (Kim et al., 2023). Other approaches exploit the representation hierarchy: EMO aligns token-level relations and hidden states across tokenizers, while TALAS anchors upper student layers to the teacher endpoint and self-distills lower layers (Truong et al., 2025; Dao et al., 2026). MoL replaces rigid layer pairing with mixtures of teacher layers, and feature-alignment distillation transfers intermediate states and attention (Vu et al., 2026; Wang et al., 2024). Prior analyses also identify overfitting in intermediate-layer distillation and limited sensitivity to layer selection (Ko et al., 2023; Yu et al., 2025). GATE-KD studies the complementary terminal-only regime using cached final teacher embeddings.

Dimension mismatch is commonly handled by mapping one representation space into the other. HPD learns projection layers for compact sentence representations, while broader studies show that projector parameterization and normalization can materially affect student optimization (Zhao et al., 2022; Miles & Mikolajczyk, 2024; Chen et al., 2023). Orthogonally constrained maps preserve intra-batch similarity, whereas TCS uses PCA coordinates extracted from cached teacher features (Miles et al., 2024; Zhou et al., 2025). Representation-similarity work more generally treats orthogonal transformations as natural equivalences because they preserve inner products and distances, with Procrustes providing the corresponding alignment (Kornblith et al., 2019; Williams et al., 2021; Maystre et al., 2025). Recent work also formalizes supervision over representation equivalence classes (Han, 2026). Most closely related to our geometric motivation, Bhattarai et al. (2025) show that unconstrained projected matching can achieve zero feature loss without preserving teacher Gram geometry, whereas Procrustes alignment avoids this shortcut under their assumptions. GATE-KD uses Procrustes as an epoch-wise coordinate-gauge update on a fixed PCA subspace rather than as a trainable projector.

Coordinate-free objectives provide a complementary route to representation structure. Relational, similarity-preserving, and contrastive distillation match pairwise distances, Gram structure, or positive–negative relations (Park et al., 2019; Tung & Mori, 2019; Tian et al., 2020). Persistent homology extends this view to multiscale properties of a point cloud. G-CRD studies local and global graph-representation topology, while JDCEmb and DICE combine distillation with geometry-shaping objectives in dialogue and diffusion settings (Joshi et al., 2024; Burykina et al., 2025; Zhou et al., 2026). More directly, TopKD transfers feature topology through approximate persistence images, and topological-guidance distillation uses persistence features from multiple teachers (Kim et al., 2024; Jeon et al., 2024). Topologically regularized data embeddings provide a broader framework for imposing prior topology on learned low-dimensional representations (Heiter et al., 2023). GATE-KD does not use topology as a surrogate for full representation geometry or sample-wise supervision. Its native-space regularizer matches sorted degree-zero persistence death times, requiring neither shared coordinates nor equal ambient dimension, while the explicit endpoint interface supplies the primary correspondence signal.

# 3 Method

Let $f_T : \mathcal{X} \to \mathbb{R}^{d_T}$ be a frozen teacher and $f_S : \mathcal{X} \to \mathbb{R}^{d_S}$ a student with $d_T > d_S$. We run the teacher once over an unlabeled corpus $\{x_i\}_{i=1}^N$ and cache its normalized final sentence embeddings

$$\mathbf{t}_i = \mathrm{norm}(f_T(x_i)) \in \mathbb{S}^{d_T-1}, \tag{1}$$

where $\mathrm{norm}(\mathbf{x}) = \mathbf{x}/\|\mathbf{x}\|_2$. The corresponding student embedding is $\mathbf{z}_i = \mathrm{norm}(\mathrm{Pool}(h_i^{(L)})) \in \mathbb{S}^{d_S-1}$. GATE-KD primarily transfers teacher knowledge through an extrinsic interface that expresses each teacher endpoint in coordinates the student can match pointwise. It augments this interface with an intrinsic regularizer that requires neither shared coordinates nor equal ambient dimension.

## 3.1 Extrinsic Endpoint Interface

**Subspace selection.** Let $\mu = N^{-1}\sum_i \mathbf{t}_i$. We fit the leading $d_S$ principal directions of the centered cache and collect them as $P_{\mathrm{PCA}} \in \mathbb{R}^{d_T \times d_S}$, with $P_{\mathrm{PCA}}^\top P_{\mathrm{PCA}} = I_{d_S}$. Centering is used to estimate the principal directions; in the evaluated recipe, the cached vectors are not centered when the projection is applied. The normalized coordinates before gauge alignment are

$$\mathbf{y}_i = \mathrm{norm}(\mathbf{t}_i P_{\mathrm{PCA}}) \in \mathbb{S}^{d_S-1}. \tag{2}$$

The PCA subspace is fitted once on the full teacher cache and remains fixed throughout training. Centering the cache to estimate covariance directions and subtracting the mean when applying the interface are distinct operations: the former selects a subspace, whereas the latter would make the target transform affine and alter its angular geometry after row normalization. We use only the former, preserving a linear endpoint interface. Appendix H evaluates both choices together with whitening and random-subspace controls.

**Gauge alignment.** The basis used to express the retained subspace affects a pointwise loss even though it does not affect the geometry within that subspace. We resolve this coordinate ambiguity with orthogonal Procrustes alignment (Schönemann, 1966; Bhattarai et al., 2025). A calibration subset $\mathcal{C}$ is selected before training and reused unchanged. Let $Y_{\mathcal{C}}$ contain its projected teacher coordinates and let $Z_{\mathcal{C}}^{(e)}$ contain the current student embeddings at the start of epoch $e$. We compute

$$R^{(e)} = \underset{R \in O(d_S)}{\arg\min} \left\| Y_{\mathcal{C}} R - Z_{\mathcal{C}}^{(e)} \right\|_F^2 = UV^\top, \quad U\Sigma V^\top = \mathrm{svd}\Big(Y_{\mathcal{C}}^\top Z_{\mathcal{C}}^{(e)}\Big). \tag{3}$$

The initialized student determines $R^{(0)}$. After each non-final epoch, we re-encode the same calibration sentences and refit the gauge for the following epoch. The target used within epoch $e$ is therefore

$$\boldsymbol{\tau}_i^{(e)} = \mathbf{y}_i R^{(e)}. \tag{4}$$

Each target is fixed within an epoch. The refit is a closed-form coordinate update rather than a learned projector.

**Relation to the endpoint objective.** Because the rows of $Y_{\mathcal{C}}$ and $Z_{\mathcal{C}}^{(e)}$ have unit norm, their cosine endpoint objective satisfies

$$\frac{1}{|\mathcal{C}|}\sum_{i\in\mathcal{C}}\left(1-\left\langle \mathbf{z}_i^{(e)}, \mathbf{y}_i R\right\rangle\right) = \frac{1}{2|\mathcal{C}|}\left\|Z_{\mathcal{C}}^{(e)} - Y_{\mathcal{C}}R\right\|_F^2. \tag{5}$$

Equation 3 therefore exactly minimizes the calibration endpoint loss over $R$ at each refit boundary with the student fixed.

**Gauge invariance.** For every orthogonal $R$,

$$(YR)(YR)^\top = YY^\top. \tag{6}$$

Consequently, refitting the gauge leaves all pairwise cosines, chord distances, and persistent-homology signatures of the projected teacher targets unchanged. PCA determines which teacher subspace survives the width bottleneck, while Procrustes determines how that fixed geometry is presented to the evolving student.

**Corollary** (Monotone gauge refit)**.** *With the student fixed, every gauge refit weakly decreases the calibration endpoint loss, while leaving the $H_0$ regularizer unchanged.*

*Proof.* Equation 3 minimizes the endpoint loss over all $R \in O(d_S)$, so its value cannot exceed that obtained with the previous gauge. The $H_0$ regularizer depends only on the fixed native teacher and student embeddings and is independent of $R$. □

### 3.2 Native-Space Topological Regularization

The endpoint interface can reproduce only the geometry retained after projection. Although $P_{\mathrm{PCA}}$ has orthonormal columns, $P_{\mathrm{PCA}}P_{\mathrm{PCA}}^\top$ is a rank-$d_S$ projector, so the cosine Gram matrix of the normalized coordinates in Equation 2 need not equal that of the original teacher embeddings. We therefore add a structural regularizer defined directly in the teacher's native space.

This loss of geometry is unavoidable whenever the normalized teacher cache has rank above the student width.

**Proposition 1** (Rank-relaxed lower bound on Gram distortion)**.** *Let $T \in \mathbb{R}^{N\times d_T}$ collect the normalized teacher cache embeddings, with singular values $\sigma_1(T) \geq \cdots \geq \sigma_r(T) > 0$. For every $Z \in \mathbb{R}^{N\times d_S}$ collecting student embeddings of the same $N$ sentences,*

$$\left\|TT^\top - ZZ^\top\right\|_F^2 \geq \sum_{j>d_S}\sigma_j(T)^4. \tag{7}$$

*Thus exact preservation of the teacher cosine Gram matrix is impossible when $\mathrm{rank}(T) > d_S$. This relaxation ranges over arbitrary rank-$d_S$ Gram approximations and need not be tight under the student's positive-semidefinite and unit-diagonal constraints.*

*Proof.* The eigenvalues of $TT^\top$ are $\sigma_j(T)^2$, whereas $\mathrm{rank}(ZZ^\top) \leq d_S$. The Eckart–Young theorem therefore lower-bounds the squared error of any such rank-$d_S$ approximation by the squared tail eigenvalues (Eckart & Young, 1936). □

For a minibatch of $B$ normalized vectors, we use the chord distance

$$d(\mathbf{u},\mathbf{v}) = \sqrt{2 - 2\langle \mathbf{u},\mathbf{v}\rangle}. \tag{8}$$

The $B-1$ finite death times of degree-zero Vietoris–Rips persistence are the edge weights of the point cloud's minimum spanning tree (Hofer et al., 2020). Let $q^S_{(1)} \leq \cdots \leq q^S_{(B-1)}$ and $q^T_{(1)} \leq \cdots \leq q^T_{(B-1)}$ denote the sorted death times of the student embeddings $\{\mathbf{z}_i\}$ and the original teacher embeddings $\{\mathbf{t}_i\}$. We define

$$\mathcal{L}_{H_0} = \frac{1}{B-1} \sum_{j=1}^{B-1} \left( q^S_{(j)} - q^T_{(j)} \right)^2 . \tag{9}$$

For a precise interpretation, define the empirical distributions of finite connectivity scales

$$\nu_S = \frac{1}{B-1} \sum_{j=1}^{B-1} \delta_{q^S_{(j)}}, \qquad \nu_T = \frac{1}{B-1} \sum_{j=1}^{B-1} \delta_{q^T_{(j)}}, \tag{10}$$

where $\delta_a$ denotes a Dirac mass at $a$.

**Proposition 2** (Transport interpretation and stability)**.** *The $H_0$ loss equals the squared one-dimensional Wasserstein distance between the empirical death-scale measures,*

$$\mathcal{L}_{H_0} = W_2^2(\nu_S, \nu_T). \tag{11}$$

*Moreover, if the two distance matrices on the corresponding batch satisfy*

$$\epsilon = \max_{i,j} \left| d(\mathbf{z}_i, \mathbf{z}_j) - d(\mathbf{t}_i, \mathbf{t}_j) \right|, \tag{12}$$

*then $\mathcal{L}_{H_0} \leq \epsilon^2$.*

*Proof.* Optimal transport on the real line pairs equally weighted samples in sorted order, which gives Equation 11 (Peyré & Cuturi, 2019). Under a uniform edge perturbation of at most $\epsilon$, every threshold graph at scale $a$ in either metric is contained in the other at scale $a + \epsilon$. Kruskal's characterization of the ordered minimum-spanning-tree edges then gives $|q^S_{(j)} - q^T_{(j)}| \leq \epsilon$ for every $j$, and averaging their squared differences proves the bound. This is the degree-zero finite-space case of Vietoris–Rips persistence stability (Chazal et al., 2014). $\square$

The teacher diagram is computed from $\mathbf{t}_i \in \mathbb{R}^{d_T}$ rather than $\mathbf{y}_i$ or $\boldsymbol{\tau}_i^{(e)}$. The loss therefore needs neither a projection nor a shared orientation. It is piecewise differentiable through the selected student edge lengths (Hofer et al., 2019). The stability result is one-way. Equation 9 matches the multiset of $H_0$ death times but does not impose correspondence between individual minimum-spanning-tree edges, recover the full metric geometry, or preserve higher-dimensional topological features.

### 3.3 Objective and Alternating Optimization

The endpoint loss for a training minibatch is

$$\mathcal{L}^{(e)}_{\text{end}} = \frac{1}{B} \sum_{i=1}^{B} \left( 1 - \left\langle \mathbf{z}_i, \boldsymbol{\tau}_i^{(e)} \right\rangle \right). \tag{13}$$

GATE-KD optimizes the two-term objective

$$\mathcal{L}^{(e)} = \mathcal{L}^{(e)}_{\text{end}} + \lambda_{\text{topo}} \mathcal{L}_{H_0}. \tag{14}$$

Here, $\mathcal{L}_{H_0}$ is evaluated on the same minibatch using the student embeddings and the corresponding unprojected teacher embeddings.

Optimization alternates between stochastic updates of the student parameters and exact updates of the coordinate gauge. During epoch $e$, $R^{(e)}$ is held fixed while the student is optimized with Equation 14. After epoch $e$, unless it is the final epoch, the updated student re-encodes $\mathcal{C}$ and Equation 3 produces $R^{(e+1)}$ for the next epoch. The $H_0$ term is unaffected by these gauge updates because it is computed from the native teacher embeddings.

Teacher embeddings and the PCA subspace are computed once before student optimization. Ordinary training steps use the cached teacher embeddings and require no online teacher forward pass. Each gauge update adds one student pass over the calibration set and an SVD of a $d_S \times d_S$ matrix.

# 4 Experiments

## 4.1 Experimental Setup

**Teacher–student configurations.** We adopt the three teacher–student pairs evaluated by TALAS (Dao et al., 2026). They comprise Qwen3-Embedding-4B $\rightarrow$ BERT-base, BGE-M3 $\rightarrow$ MiniLMv2-L6-H768, and Qwen3-Embedding-0.6B $\rightarrow$ MiniLMv2-L6-H384 (Devlin et al., 2019; Wang et al., 2021; Chen et al., 2024; Zhang et al., 2025). These settings span students from 22M to 109M parameters and embedding bottlenecks of $2560 \rightarrow 768$, $1024 \rightarrow 768$, and $1024 \rightarrow 384$, respectively. The Qwen teachers use last-token pooling, whereas BGE-M3 uses CLS pooling.

**Distillation corpus.** We construct a shared distillation corpus by sampling approximately 5K unlabeled training texts from each of Emotion, WiC, and STS-B. No labels or task annotations are used during distillation, and every method uses the same corpus. We refer to these three benchmarks as source tasks and the remaining six as non-source tasks. Tables retain the compact labels IOD and OOD, respectively; here OOD means only that a benchmark did not supply distillation text, not an unrestricted distribution shift or corpus-disjoint pretraining regime.

**Baselines.** We compare GATE-KD with methods covering complementary transfer mechanisms. RKD matches inter-example relations (Park et al., 2019), while Stella uses a multi-stage embedding-distillation recipe (Zhang et al., 2024a). CDM and DSKD address cross-tokenizer alignment and dual-space transfer, respectively (Chen et al., 2025; Zhang et al., 2024b). EMO combines token-level relations with hidden-state alignment (Truong et al., 2025), and TALAS combines teacher anchoring with hierarchy-based self-distillation (Dao et al., 2026). Method-specific training hyperparameters are reported in Appendix A.

**Evaluation protocol.** We evaluate frozen student embeddings on nine benchmarks from three task families. For classification on Banking77, Emotion, and Tweet, we fit a logistic-regression probe on each benchmark's training split and report macro-F1. For pair classification on MRPC, SciTail, and WiC, we map cosine similarity to $[0, 1]$ and report average precision. For semantic textual similarity on SICK, STS12, and STS-B, we report Spearman correlation with the gold scores. All metrics are multiplied by 100. Avg. is the unweighted mean of the nine primary scores, while Avg. IOD and Avg. OOD average the three in-domain and six out-of-domain benchmarks, respectively. We report means and sample standard deviations over three seeds.

## 4.2 Main Results

Table 1 compares GATE-KD with six distillation baselines across nine benchmarks and three teacher–student configurations. GATE-KD obtains the numerically highest overall average in all three configurations, improving over the next-best baseline by 0.38, 0.48, and 1.59 points in configurations (a), (b), and (c), respectively. The advantage is largest for the 22M-parameter student and therefore does not diminish in the most capacity-constrained configuration.

The task-level results show that this advantage is broad but not uniform. GATE-KD ranks first on 18 of the 27 individual benchmark comparisons, with its most consistent results on classification and semantic textual similarity. Other methods remain stronger on some pair-classification benchmarks, including MRPC and WiC. In the largest capacity-gap configuration, GATE-KD essentially matches TALAS in-domain, with 72.04 versus 72.19 Avg. IOD, while improving Avg. OOD from 79.85 to 80.49. The aggregate gain in this configuration is therefore driven by OOD performance rather than higher scores on the datasets supplying the distillation text.

## 4.3 Measured Optimization Cost

Table 2 reports the cost of student optimization under the same execution protocol on NVIDIA H200 GPUs. Step time and throughput exclude the first ten warm-up steps. GPU optimization time sums CUDA-event durations for the minibatch forward, loss, backward, and optimizer update. It therefore includes the topology computation performed inside GATE-KD's training steps, but excludes model and tokenizer loading, data preprocessing, final evaluation, teacher-cache construction, the one-time

Table 1: Results across nine sentence-level benchmarks. Distilled-student scores are mean ± sample standard deviation over three seeds; best and second-best results are bold and underlined.

| Method | Classification (F1) | | | Pair Classification (AP) | | | STS (Spearman) | | | Averages | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Banking77 | Tweet | Emotion | MRPC | SciTail | WiC | SICK | STS12 | STS-B | IOD | OOD | All |
| *(a) Qwen3-Embedding-4B → BERT-base (109M)* | | | | | | | | | | | | |
| Teacher | 93.26 | 71.04 | 69.08 | 84.80 | 91.77 | 66.79 | 83.43 | 82.55 | 86.87 | 74.25 | 84.48 | 81.07 |
| Student base | 84.33 | 67.08 | 47.79 | 77.36 | 67.74 | 59.22 | 42.43 | 21.54 | 20.29 | 42.43 | 60.08 | 54.20 |
| RKD | 89.76 ± 0.11 | 70.90 ± 0.31 | 59.98 ± 0.05 | 81.94 ± 0.26 | 75.59 ± 0.13 | 69.02 ± 0.18 | 72.93 ± 0.07 | 67.84 ± 0.18 | 73.72 ± 0.08 | 67.57 ± 0.05 | 76.49 ± 0.07 | 73.52 ± 0.06 |
| Stella | 88.48 ± 0.30 | 68.94 ± 0.17 | 53.69 ± 0.50 | 83.95 ± 0.10 | 75.39 ± 0.54 | **71.93** ± 0.09 | 69.87 ± 0.64 | 64.39 ± 0.21 | 70.86 ± 0.36 | 65.49 ± 0.17 | 75.17 ± 0.13 | 71.95 ± 0.06 |
| CDM | 88.13 ± 0.23 | 69.03 ± 0.20 | 52.83 ± 0.50 | 83.00 ± 0.06 | 77.10 ± 0.27 | 71.84 ± 0.09 | 67.89 ± 0.37 | 61.06 ± 0.37 | 69.19 ± 0.42 | 64.62 ± 0.28 | 74.37 ± 0.12 | 71.12 ± 0.18 |
| DSKD | 87.97 ± 0.16 | 68.90 ± 0.28 | 53.07 ± 0.31 | 83.03 ± 0.15 | 76.98 ± 0.59 | 71.86 ± 0.08 | 67.77 ± 0.66 | 61.11 ± 0.26 | 69.55 ± 0.12 | 64.83 ± 0.12 | 74.29 ± 0.14 | 71.14 ± 0.10 |
| EMO | 86.89 ± 0.07 | 68.99 ± 0.11 | 52.02 ± 1.09 | 79.74 ± 0.12 | 76.37 ± 0.35 | 69.62 ± 0.16 | 67.89 ± 0.09 | 56.46 ± 0.47 | 64.54 ± 0.50 | 62.06 ± 0.49 | 72.72 ± 0.15 | 69.17 ± 0.24 |
| TALAS | 90.53 ± 0.09 | 73.77 ± 0.21 | 68.47 ± 0.34 | **85.38** ± 0.05 | **81.83** ± 0.21 | 69.59 ± 0.14 | 76.79 ± 0.06 | 70.79 ± 0.19 | **78.50** ± 0.08 | **72.19** ± 0.07 | 79.85 ± 0.07 | 77.29 ± 0.04 |
| **GATE-KD** | **90.60** ± 0.21 | **73.86** ± 0.10 | **69.48** ± 0.22 | 85.31 ± 0.11 | 80.23 ± 0.13 | 68.19 ± 0.12 | **77.80** ± 0.14 | **75.15** ± 0.07 | 78.44 ± 0.16 | 72.04 ± 0.08 | **80.49** ± 0.03 | **77.67** ± 0.03 |
| *(b) BGE-M3 → MiniLMv2-H768 (66M)* | | | | | | | | | | | | |
| Teacher | 93.49 | 73.75 | 69.03 | 85.80 | 91.87 | 61.46 | 79.18 | 78.74 | 84.86 | 71.78 | 83.81 | 79.80 |
| Student base | 81.18 | 71.30 | 48.85 | 79.66 | 65.06 | 60.89 | 49.46 | 26.76 | 24.12 | 44.62 | 62.24 | 56.36 |
| RKD | 89.14 ± 0.09 | 73.29 ± 0.09 | 62.23 ± 0.27 | 84.35 ± 0.06 | 78.74 ± 0.11 | 65.02 ± 0.10 | 73.20 ± 0.19 | 65.02 ± 0.29 | 71.97 ± 0.38 | 66.41 ± 0.07 | 77.29 ± 0.06 | 73.66 ± 0.04 |
| Stella | 86.82 ± 0.28 | 70.06 ± 0.12 | 56.61 ± 0.74 | 83.02 ± 0.12 | 74.63 ± 0.12 | **69.72** ± 0.05 | 69.08 ± 0.20 | 63.14 ± 0.28 | 67.10 ± 0.10 | 64.47 ± 0.27 | 74.46 ± 0.05 | 71.13 ± 0.11 |
| CDM | 87.82 ± 0.19 | 69.99 ± 0.09 | 58.32 ± 0.51 | 83.44 ± 0.07 | 75.56 ± 0.40 | 69.70 ± 0.02 | 65.93 ± 0.16 | 60.84 ± 0.34 | 65.31 ± 0.15 | 64.45 ± 0.13 | 73.93 ± 0.06 | 70.77 ± 0.02 |
| DSKD | 87.70 ± 0.16 | 70.35 ± 0.32 | 59.71 ± 0.61 | 82.99 ± 0.18 | 75.58 ± 0.12 | 69.59 ± 0.04 | 65.36 ± 0.37 | 61.21 ± 0.31 | 65.02 ± 0.18 | 64.77 ± 0.27 | 73.87 ± 0.10 | 70.83 ± 0.11 |
| EMO | 87.54 ± 0.18 | 71.03 ± 0.14 | 58.52 ± 0.30 | 82.46 ± 0.18 | 74.73 ± 0.58 | 69.20 ± 0.22 | 64.71 ± 0.55 | 56.89 ± 0.27 | 61.25 ± 0.03 | 62.99 ± 0.19 | 72.89 ± 0.10 | 69.59 ± 0.12 |
| TALAS | 89.83 ± 0.09 | **74.70** ± 0.09 | 61.65 ± 0.23 | **86.62** ± 0.05 | 80.85 ± 0.22 | 66.72 ± 0.07 | 75.40 ± 0.10 | 67.08 ± 0.31 | 77.08 ± 0.01 | 68.48 ± 0.07 | 79.08 ± 0.08 | 75.55 ± 0.06 |
| **GATE-KD** | **90.17** ± 0.05 | 73.74 ± 0.24 | **64.83** ± 0.72 | 86.42 ± 0.09 | **80.97** ± 0.08 | 65.47 ± 0.09 | **76.77** ± 0.03 | **68.27** ± 0.04 | **77.63** ± 0.12 | **69.31** ± 0.21 | **79.39** ± 0.06 | **76.03** ± 0.11 |
| *(c) Qwen3-Embedding-0.6B → MiniLMv2-H384 (22M)* | | | | | | | | | | | | |
| Teacher | 92.51 | 71.28 | 65.80 | 83.38 | 90.09 | 66.84 | 80.65 | 77.11 | 84.59 | 72.41 | 82.50 | 79.14 |
| Student base | 67.03 | 69.45 | 42.90 | 78.39 | 64.55 | 59.58 | 47.60 | 21.96 | 22.08 | 41.52 | 58.16 | 52.62 |
| RKD | 86.76 ± 0.07 | 69.87 ± 0.21 | 56.31 ± 0.35 | 81.70 ± 0.03 | 76.03 ± 0.20 | 68.56 ± 0.22 | 69.21 ± 0.18 | 64.26 ± 0.19 | 70.25 ± 0.21 | 65.04 ± 0.20 | 74.64 ± 0.09 | 71.44 ± 0.10 |
| Stella | 84.04 ± 0.04 | 68.73 ± 0.44 | 52.41 ± 0.28 | 81.50 ± 0.12 | 72.26 ± 0.32 | **68.65** ± 0.16 | 64.64 ± 0.19 | 60.59 ± 0.12 | 64.27 ± 0.20 | 61.78 ± 0.13 | 71.96 ± 0.13 | 68.57 ± 0.10 |
| CDM | 84.74 ± 0.21 | 69.52 ± 0.20 | 53.76 ± 0.95 | 81.15 ± 0.14 | 72.86 ± 0.30 | 68.29 ± 0.11 | 63.98 ± 0.36 | 60.43 ± 0.22 | 63.52 ± 0.24 | 61.86 ± 0.43 | 72.11 ± 0.19 | 68.69 ± 0.27 |
| DSKD | 84.76 ± 0.10 | 69.56 ± 0.20 | 53.14 ± 0.29 | 81.15 ± 0.01 | 72.81 ± 0.09 | 68.33 ± 0.04 | 63.78 ± 0.06 | 60.24 ± 0.19 | 63.53 ± 0.18 | 61.67 ± 0.03 | 72.05 ± 0.06 | 68.59 ± 0.04 |
| EMO | 83.54 ± 0.31 | 69.31 ± 0.06 | 54.48 ± 0.27 | 80.87 ± 0.13 | 72.00 ± 0.16 | 67.89 ± 0.09 | 62.19 ± 0.16 | 57.04 ± 0.17 | 61.24 ± 0.23 | 61.20 ± 0.09 | 70.82 ± 0.07 | 67.62 ± 0.02 |
| TALAS | 86.39 ± 0.13 | 71.81 ± 0.11 | 58.40 ± 0.30 | **84.25** ± 0.03 | 79.29 ± 0.11 | 66.82 ± 0.02 | 70.42 ± 0.10 | 68.08 ± 0.22 | 73.34 ± 0.10 | 66.19 ± 0.11 | 76.71 ± 0.04 | 73.20 ± 0.06 |
| **GATE-KD** | **87.65** ± 0.06 | **72.32** ± 0.24 | **63.04** ± 0.28 | 84.20 ± 0.06 | **81.60** ± 0.05 | 68.07 ± 0.12 | **72.48** ± 0.08 | **68.96** ± 0.18 | **74.76** ± 0.14 | **68.62** ± 0.21 | **77.87** ± 0.02 | **74.79** ± 0.06 |

PCA fit, and the four student re-encoding passes used for gauge refitting. The table measures the cost of the optimization loop rather than end-to-end wall-clock cost.

Teacher access also differs across method families. GATE-KD, RKD, Stella, and TALAS read normalized final teacher embeddings from the same reusable cache and do not execute the teacher during student optimization. CDM, DSKD, and EMO execute the teacher online; EMO additionally consumes teacher hidden states and attention. Charging a shared cache entirely to the first cached method would be arbitrary, while excluding it understates the cost of a standalone run. We therefore report the cache-independent optimization measurement and do not interpret it as total training cost.

GATE-KD uses 26%, 31%, and 38% of TALAS's measured GPU optimization time in configurations (a), (b), and (c), respectively, while also attaining higher overall scores. The improvement is primarily in throughput rather than peak memory allocation, for which TALAS remains lower across the three configurations. A controlled end-to-end comparison would additionally require isolated runs that separately time teacher encoding, cache I/O, PCA and gauge updates, preprocessing, and evaluation; the present measurements support only the stated optimization-loop comparison, not an end-to-end efficiency claim.

Table 2: Student-optimization cost on NVIDIA H200 GPUs. Values are mean ± sample standard deviation over three seeds; best and second-best are bold and underlined. Timings cover minibatch forward, loss, backward, and optimizer updates only.

| Method | Avg. ↑ | ms/step ↓ | samples/s ↑ | GPU opt. min ↓ | peak GiB ↓ |
|---|---|---|---|---|---|
| *(a) Qwen3-Embedding-4B → BERT-base (109M)* | | | | | |
| RKD | 73.52 ± 0.06 | 119.7 ± 18.3 | 1081.0 ± 179.8 | 1.20 ± 0.18 | 1.72 ± 0.00 |
| Stella | 71.95 ± 0.06 | 130.7 ± 8.1 | 975.9 ± 58.8 | 1.27 ± 0.08 | 1.78 ± 0.01 |
| CDM | 71.12 ± 0.18 | 340.4 ± 13.6 | 374.2 ± 15.3 | 3.38 ± 0.12 | 9.31 ± 0.01 |
| DSKD | 71.14 ± 0.10 | 290.9 ± 9.0 | 437.7 ± 13.3 | 2.88 ± 0.09 | 9.45 ± 0.00 |
| EMO | 69.17 ± 0.24 | 721.1 ± 12.9 | 176.5 ± 3.1 | 7.02 ± 0.12 | 9.30 ± 0.00 |
| TALAS | 77.29 ± 0.04 | 329.0 ± 12.5 | 387.1 ± 15.0 | 3.23 ± 0.11 | **1.43** ± 0.00 |
| **GATE-KD** | **77.67** ± 0.03 | **83.4** ± 3.8 | **1527.2** ± 70.8 | **0.84** ± 0.04 | 1.72 ± 0.00 |
| *(b) BGE-M3 → MiniLMv2-H768 (66M)* | | | | | |
| RKD | 73.66 ± 0.04 | 77.5 ± 8.8 | 1656.4 ± 201.2 | 0.79 ± 0.09 | 1.07 ± 0.00 |
| Stella | 71.13 ± 0.11 | 152.8 ± 30.8 | 859.1 ± 195.6 | 1.49 ± 0.30 | 1.09 ± 0.00 |
| CDM | 70.77 ± 0.02 | 127.5 ± 13.8 | 1006.4 ± 115.8 | 1.30 ± 0.12 | 2.15 ± 0.01 |
| DSKD | 70.83 ± 0.11 | 115.0 ± 21.9 | 1137.2 ± 241.0 | 1.14 ± 0.20 | 2.22 ± 0.01 |
| EMO | 69.59 ± 0.12 | 394.6 ± 33.4 | 324.0 ± 28.3 | 3.87 ± 0.35 | 2.15 ± 0.01 |
| TALAS | 75.55 ± 0.06 | 184.4 ± 4.6 | 690.2 ± 17.3 | 1.82 ± 0.03 | **0.85** ± 0.00 |
| **GATE-KD** | **76.03** ± 0.11 | **56.5** ± 7.7 | **2281.6** ± 310.0 | **0.57** ± 0.07 | 1.07 ± 0.00 |
| *(c) Qwen3-Embedding-0.6B → MiniLMv2-H384 (22M)* | | | | | |
| RKD | 71.44 ± 0.10 | 64.8 ± 13.7 | 2012.3 ± 488.6 | 0.68 ± 0.14 | 0.41 ± 0.00 |
| Stella | 68.57 ± 0.10 | 150.7 ± 15.3 | 850.0 ± 81.5 | 1.47 ± 0.13 | 0.42 ± 0.00 |
| CDM | 68.69 ± 0.27 | 161.8 ± 21.6 | 796.3 ± 113.1 | 1.65 ± 0.20 | 1.53 ± 0.00 |
| DSKD | 68.59 ± 0.04 | 175.4 ± 8.2 | 726.4 ± 33.5 | 1.78 ± 0.06 | 1.56 ± 0.00 |
| EMO | 67.62 ± 0.02 | 499.3 ± 1.4 | 254.8 ± 0.7 | 4.90 ± 0.01 | 1.53 ± 0.00 |
| TALAS | 73.20 ± 0.06 | 169.8 ± 15.4 | 753.7 ± 71.4 | 1.69 ± 0.14 | **0.34** ± 0.00 |
| **GATE-KD** | **74.79** ± 0.06 | **62.9** ± 3.1 | **2025.2** ± 98.3 | **0.64** ± 0.04 | 0.41 ± 0.00 |

# 5 Controlled Analysis

We ask whether the endpoint gain comes from coordinate alignment rather than subspace selection, and whether the auxiliary $H_0$ gain is specific to its connectivity statistic. Unless stated otherwise, controls use configuration (c), whose 384-dimensional student has the strongest width bottleneck. Extended controls appear in Appendix B–I.

## 5.1 Coordinate Comparability Drives Endpoint Transfer

Table 3 holds the PCA subspace and endpoint objective fixed. In configuration (c), no alignment, a random orientation, and epoch-wise Procrustes reach 69.35, 70.37, and 74.28 Avg. Because these variants retain the same teacher geometry, the 4.93-point gain over no alignment comes from coordinate comparability rather than retained variance; the 3.91-point gap to a random rotation shows that rotation alone is insufficient.

Fixed-$R^{(0)}$ and one-refit controls separate initial comparability from tracking the evolving student; configuration (a) tests whether the mechanism persists beyond the compact setting. Learned-map and subspace controls appear in Appendix B.

Table 3: **Gauge mechanism under the full base objective.** Every row uses the same PCA subspace, student, corpus, optimizer, and training schedule, including the same $H_0$ term; only the orientation and its update schedule change. Values are mean $\pm$ sample standard deviation over three seeds.

| Gauge | Data fitted | Updates | Config. (a) | Config. (c) |
|---|---|---|---|---|
| None | No | 0 | **[run]** | **[run]** |
| Random orthogonal | No | 0 | **[run]** | **[run]** |
| Fixed $R^{(0)}$ | Calibration set | 0 | **[run]** | **[run]** |
| One refit | Calibration set | 1 | **[run]** | **[run]** |
| Epoch-wise Procrustes | Calibration set | 4 | **[run]** | **[run]** |

### 5.2 What Does the Topological Term Add?

Within the topology-weight sweep, endpoint-only supervision reaches 74.21 Avg.; native-space $H_0$ at $\lambda_{\text{topo}} = 0.5$ reaches 74.79, a 0.58-point refinement. Table 4 compares it with matched pairwise, local-scale, teacher-MST, and Gram objectives while holding the endpoint interface and optimization protocol fixed.

Table 4: **Matched structural controls in configuration (c).** All variants retain the same PCA–Procrustes endpoint objective and differ only in the auxiliary structural statistic. "Correspondence" indicates whether the loss compares the same indexed sample pairs. Optimization time is measured over the same forward–backward scope as Table 2.

| Auxiliary term | Matched object | Correspondence | Avg. ↑ | Δ endpoint | ms/step ↓ |
|---|---|---|---|---|---|
| None | — | Pointwise endpoint | 74.21$\pm$ 0.03 | — | **[run]** |
| Sorted spanning path | Distance distribution | No | **[run]** | **[run]** | **[run]** |
| kNN distribution | Local-scale distribution | No | **[run]** | **[run]** | **[run]** |
| Teacher-MST edges | $B - 1$ selected distances | Yes | **[run]** | **[run]** | **[run]** |
| Native Gram matching | All pairwise cosines | Yes | **[run]** | **[run]** | **[run]** |
| $H_0$ | Sorted MST death scales | No | **74.79**$\pm$ 0.06 | +0.58 | **[run]** |

The sorted spanning-path and kNN rows test whether any distance distribution suffices; teacher-MST and Gram rows restore sample correspondence; $H_0$ retains only sorted connectivity scales. A topology-specific interpretation therefore requires $H_0$ to improve over these controls. In a separate suite, Appendix C shows that $H_0$ alone reaches 64.22, so connectivity does not replace pointwise correspondence. Weight and calibration sensitivity appear in Appendix D.

## 6 Conclusion

GATE-KD treats cross-dimensional embedding distillation as an interface problem, using a fixed PCA subspace and closed-form gauge alignment with native-space $H_0$ regularization. Across three teacher–student configurations, this final-embedding-only formulation obtains the numerically highest aggregate scores among the evaluated distilled students while reducing measured optimization-loop time relative to TALAS. The analyses identify coordinate comparability as the primary source of improvement and $H_0$ as a complementary regularizer. The current sentence-level objective does not fully preserve the teacher's zero-shot retrieval capability, leaving broader retrieval transfer for future work. In addition, the current corpus uses training text from three downstream source benchmarks, so its source-task results measure label-free task adaptation rather than corpus-disjoint transfer. The conclusions are therefore limited to the task-adapted sentence-level regime evaluated here.

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# A TRAINING DETAILS

All main-comparison runs use a maximum sequence length of 256 and seeds 42, 43, and 44. We fix the batch size at 128 and the total training duration at five epochs for every method; Stella's two stages comprise two and three epochs. Learning rate is the only tuned optimization hyperparameter and is selected for every method using validation data. The selected rate is then fixed for the three reported seeds, and we evaluate the final-epoch checkpoint once on the test splits. No test score is used for learning-rate or checkpoint selection. For GATE-KD, $\lambda_{\mathrm{topo}} = 0.5$, the $H_0$ cloud is the current training batch, the gauge is refitted after each non-final epoch, and calibration uses the full distillation corpus. These settings are fixed before test evaluation; the sensitivity sweeps are diagnostic rather than part of model selection. Table 5 lists the selected learning rates.

Table 5: Main-experiment training hyperparameters.

| Method | Epochs | Batch size | Learning rate |
|---|---|---|---|
| RKD | 5 | 128 | $7 \times 10^{-5}$ |
| Stella | $2+3$ | 128 | $5 \times 10^{-5}$ |
| CDM | 5 | 128 | $2 \times 10^{-5}$ |
| DSKD | 5 | 128 | $2 \times 10^{-5}$ |
| EMO | 5 | 128 | $1 \times 10^{-5}$ |
| TALAS | 5 | 128 | $2 \times 10^{-5}$ |
| GATE-KD | 5 | 128 | $7 \times 10^{-5}$ |

# B COMPLETE ENDPOINT-INTERFACE CONTROLS

Table 6 complements the gauge-schedule analysis in Table 3 by varying the broader endpoint interface. All rows use endpoint-only supervision in configuration (c). The learned controls use a single bias-free linear map, either from teacher to student width or from student to teacher width, followed by row normalization. The map is trained jointly with the student under the same AdamW schedule and is discarded at inference. Thus these rows compare supervision interfaces rather than deployed model capacities.

Table 6: Endpoint-interface ablation with endpoint-only supervision in configuration (c). Scores are mean $\pm$ sample standard deviation over three seeds; random controls share one map draw. Best values are bold.

| Target map | Gauge | Retained energy ↑ | Avg. IOD ↑ | Avg. OOD ↑ | Avg. ↑ |
|---|---|---|---|---|---|
| Learned $T \to S$ | — | N/A | 54.34$\pm$ 0.68 | 65.02$\pm$ 0.25 | 61.46$\pm$ 0.39 |
| Learned $S \to T$ | — | N/A | 62.54$\pm$ 0.36 | 72.72$\pm$ 0.31 | 69.33$\pm$ 0.32 |
| PCA | None | **0.928** | 62.49$\pm$ 0.19 | 72.78$\pm$ 0.07 | 69.35$\pm$ 0.04 |
| PCA | Random orthogonal | **0.928** | 63.68$\pm$ 0.10 | 73.72$\pm$ 0.09 | 70.37$\pm$ 0.08 |
| Random subspace | None | 0.381 | 64.44$\pm$ 0.14 | 74.16$\pm$ 0.04 | 70.92$\pm$ 0.07 |
| **PCA** | **Procrustes** | **0.928** | **67.76**$\pm$ 0.05 | **77.54**$\pm$ 0.07 | **74.28**$\pm$ 0.03 |

Within the fixed PCA subspace, fitted Procrustes alignment improves over both unaligned and randomly oriented targets. Under the matched recipe evaluated here, the fixed PCA–Procrustes interface also exceeds learned linear maps in both directions. This comparison is specific to the stated map architecture and optimization protocol; it does not establish that every learned projector is inferior. Subspace selection is isolated separately in Appendix H, where every PCA and random-subspace target is paired with the same fitted gauge.

# C ENDPOINT AND TOPOLOGY SIGNAL DECOMPOSITION

Table 7 separates the endpoint and topology signals. The endpoint-only objective supplies most of the downstream gain, whereas $H_0$-only training reduces the connectivity residual without reaching the endpoint-supervised score. The combined objective treats native-space connectivity as an auxiliary constraint rather than as a substitute for sample-wise correspondence. The projected-space and fixed-gauge rows test two secondary design choices. This controlled suite uses $\lambda_{\text{topo}} = 1$ for every $H_0$-active row; its purpose is decomposition rather than reproducing the $\lambda_{\text{topo}} = 0.5$ main recipe.

Table 7: Endpoint, topology, and gauge-refit decomposition in configuration (c). Values are mean ± sample standard deviation over three seeds; $H_0$ residuals use the native teacher space. Best values are bold.

| Configuration | $H_0$ source | Gauge refit | $H_0$ residual ↓ | Avg. IOD ↑ | Avg. OOD ↑ | Avg. ↑ |
|---|---|---|---|---|---|---|
| No-teacher control | — | — | 0.3595± 0.0000 | 41.61± 0.00 | 58.10± 0.00 | 52.60± 0.00 |
| $H_0$ only | Original | — | **0.1240**± 0.0012 | 58.00± 1.01 | 67.33± 0.62 | 64.22± 0.73 |
| Endpoint only | — | On | 0.2122± 0.0004 | 67.68± 0.19 | 77.55± 0.07 | 74.26± 0.02 |
| Endpoint $+H_0$ | Projected | On | 0.1355± 0.0003 | 68.27± 0.12 | 77.72± 0.00 | 74.57± 0.04 |
| Endpoint $+H_0$ | Original | Off | 0.1309± 0.0001 | 68.03± 0.10 | 77.71± 0.03 | 74.48± 0.01 |
| Endpoint $+H_0$ | Original | On | 0.1281± 0.0003 | **68.38**± 0.18 | **77.77**± 0.03 | **74.64**± 0.04 |

## D Detailed Sensitivity Results

Table 8 varies the topology weight and gauge-calibration size in configuration (c); settings not varied within each block remain at the main recipe. Every tested nonzero topology weight improves over the endpoint-only setting, while the values from 0.5 to 1.0 differ by at most 0.24 points. Reducing the calibration set from the full corpus to 2,048 sentences changes Avg. by 0.28 points, indicating that the gauge can be estimated from a relatively small sample. We omit the earlier epoch-matched batch-size sweep because changing the batch size also changed the number of optimizer updates and therefore did not isolate batch size.

Table 8: Sensitivity results in configuration (c), reported as mean ± sample standard deviation over three seeds. Best values within each block are bold.

| Factor | Value | Avg. IOD ↑ | Avg. OOD ↑ | Avg. ↑ |
|---|---|---|---|---|
| $\lambda_{\text{topo}}$ | 0 | 67.54± 0.18 | 77.54± 0.07 | 74.21± 0.03 |
| | 0.5 | **68.62**± 0.21 | **77.87**± 0.02 | **74.79**± 0.06 |
| | 0.75 | 68.47± 0.13 | 77.84± 0.04 | 74.72± 0.04 |
| | 1 | 68.14± 0.06 | 77.75± 0.03 | 74.55± 0.03 |
| Gauge calibration | 2,048 | 68.36± 0.11 | 77.59± 0.04 | 74.51± 0.06 |
| | 4,096 | 68.52± 0.08 | 77.74± 0.05 | 74.67± 0.02 |
| | 8,192 | 68.26± 0.08 | 77.78± 0.03 | 74.61± 0.02 |
| | 14,760 | **68.62**± 0.21 | **77.87**± 0.02 | **74.79**± 0.06 |

## E Results with a Broader 100K Distillation Corpus

We repeat the complete method sweep on approximately 100K unlabeled texts drawn from the training splits of all nine sentence-level benchmarks. No labels or task annotations are used. Relative to the main comparison, this setting changes both corpus size and dataset coverage. It therefore evaluates robustness in a broader task-adapted setting rather than isolating the effect of corpus size alone.

Table 9: Results with the broader 100K distillation corpus. Distilled-student scores are mean ± sample standard deviation over three seeds; best and second-best results are bold and underlined.

| | Classification (F1) | | | Pair Classification (AP) | | | STS (Spearman) | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Method | Banking77 | Tweet | Emotion | MRPC | SciTail | WiC | SICK | STS12 | STS-B | Avg. |
| *(a) Qwen3-Embedding-4B → BERT-base (109M)* | | | | | | | | | | |
| Teacher | 93.26 | 71.04 | 69.08 | 84.80 | 91.77 | 66.79 | 83.43 | 82.55 | 86.87 | 81.07 |
| Student base | 84.33 | 67.08 | 47.79 | 77.36 | 67.74 | 59.22 | 42.43 | 21.54 | 20.29 | 54.20 |
| RKD | 92.23± 0.15 | 71.96± 0.18 | 63.79± 0.40 | 85.00± 0.04 | 80.27± 0.38 | 66.31± 0.10 | 79.32± 0.02 | 71.72± 0.16 | 77.70± 0.20 | 76.48± 0.04 |
| Stella | 90.02± 0.15 | 71.90± 0.22 | 60.14± 0.45 | 85.96± 0.05 | 78.82± 0.27 | 69.93± 0.13 | 76.53± 0.23 | 69.79± 0.12 | 77.22± 0.16 | 75.59± 0.10 |
| CDM | 89.16± 0.04 | 69.63± 0.23 | 53.74± 0.13 | 85.57± 0.08 | 71.22± 0.19 | **70.04**± 0.05 | 71.26± 0.29 | 66.97± 0.21 | 73.69± 0.38 | 72.36± 0.13 |
| DSKD | 88.87± 0.07 | 69.05± 0.09 | 54.02± 0.56 | 85.90± 0.05 | 70.02± 0.31 | 69.63± 0.21 | 71.07± 0.10 | 66.52± 0.17 | 73.52± 0.01 | 72.07± 0.03 |
| EMO | 84.96± 0.84 | 68.93± 0.65 | 54.80± 1.52 | 85.05± 0.04 | 65.75± 0.22 | 68.80± 0.13 | 62.00± 2.71 | 61.75± 1.85 | 69.54± 0.37 | 69.07± 0.89 |
| TALAS | 93.12± 0.13 | **74.67**± 0.38 | 71.42± 0.26 | **86.30**± 0.01 | 87.06± 0.12 | 68.09± 0.02 | 80.29± 0.03 | 76.34± 0.01 | 82.26± 0.05 | 79.95± 0.08 |
| **GATE-KD** | **93.47**± 0.12 | 74.18± 0.25 | **72.50**± 0.47 | 85.45± 0.08 | **87.24**± 0.12 | 66.57± 0.08 | **81.85**± 0.06 | **76.40**± 0.07 | **82.52**± 0.08 | **80.02**± 0.03 |
| *(b) BGE-M3 → MiniLMv2-H768 (66M)* | | | | | | | | | | |
| Teacher | 93.49 | 73.75 | 69.03 | 85.80 | 91.87 | 61.46 | 79.18 | 78.74 | 84.86 | 79.80 |
| Student base | 81.18 | 71.30 | 48.85 | 79.66 | 65.06 | 60.89 | 49.46 | 26.76 | 24.12 | 56.36 |
| RKD | 91.68± 0.06 | 74.57± 0.40 | 62.45± 0.75 | 85.81± 0.08 | 79.81± 0.17 | 63.57± 0.13 | 76.91± 0.05 | 69.54± 0.11 | 75.56± 0.09 | 75.54± 0.10 |
| Stella | 88.72± 0.41 | 73.43± 0.23 | 61.63± 0.29 | 86.39± 0.08 | 74.13± 0.31 | 66.98± 0.27 | 73.32± 0.07 | 68.30± 0.12 | 72.66± 0.14 | 73.95± 0.04 |
| CDM | 87.64± 0.23 | 70.82± 0.14 | 56.37± 0.57 | 85.67± 0.08 | 67.69± 0.23 | 66.89± 0.20 | 70.15± 0.48 | 66.68± 0.26 | 69.49± 0.15 | 71.27± 0.10 |
| DSKD | 86.69± 0.38 | 69.76± 0.22 | 56.98± 0.39 | 84.92± 0.07 | 66.75± 0.17 | **67.23**± 0.10 | 68.26± 0.45 | 66.28± 0.02 | 68.31± 0.35 | 70.58± 0.10 |
| EMO | 87.17± 0.59 | 71.07± 0.22 | 59.32± 0.02 | 84.25± 0.41 | 67.89± 0.92 | 66.09± 0.18 | 69.04± 0.38 | 62.26± 1.77 | 66.47± 0.40 | 70.40± 0.35 |
| TALAS | 92.44± 0.10 | 75.58± 0.15 | 65.13± 0.21 | **87.09**± 0.09 | 85.39± 0.14 | 64.39± 0.12 | 78.14± 0.06 | 72.56± 0.13 | **81.03**± 0.05 | 77.97± 0.03 |
| **GATE-KD** | **92.56**± 0.04 | **76.30**± 0.31 | **65.62**± 0.29 | 86.58± 0.05 | **87.20**± 0.06 | 63.43± 0.02 | **79.23**± 0.03 | **72.69**± 0.05 | 80.60± 0.13 | **78.24**± 0.05 |
| *(c) Qwen3-Embedding-0.6B → MiniLMv2-H384 (22M)* | | | | | | | | | | |
| Teacher | 92.51 | 71.28 | 65.80 | 83.38 | 90.09 | 66.84 | 80.65 | 77.11 | 84.59 | 79.14 |
| Student base | 67.03 | 69.45 | 42.90 | 78.39 | 64.55 | 59.58 | 47.60 | 21.96 | 22.08 | 52.62 |
| RKD | 89.74± 0.04 | 72.34± 0.22 | 57.99± 0.52 | 84.13± 0.12 | 75.20± 0.14 | **67.49**± 0.06 | 74.48± 0.03 | 68.12± 0.15 | 72.55± 0.14 | 73.56± 0.09 |
| Stella | 85.74± 0.22 | 70.25± 0.07 | 53.48± 0.28 | 84.43± 0.04 | 71.23± 0.05 | 66.51± 0.14 | 70.96± 0.28 | 63.02± 0.04 | 67.42± 0.24 | 70.34± 0.02 |
| CDM | 85.18± 0.08 | 69.14± 0.07 | 52.30± 0.26 | 84.02± 0.04 | 68.24± 0.23 | 66.57± 0.29 | 69.15± 0.13 | 63.05± 0.06 | 66.55± 0.10 | 69.36± 0.08 |
| DSKD | 85.04± 0.14 | 69.21± 0.13 | 52.85± 0.36 | 84.34± 0.10 | 67.33± 0.24 | 65.98± 0.12 | 69.01± 0.13 | 63.46± 0.21 | 66.42± 0.09 | 69.29± 0.09 |
| EMO | 85.08± 0.17 | 68.11± 0.10 | 53.96± 0.60 | 84.38± 0.16 | 65.48± 0.16 | 65.23± 0.17 | 68.61± 0.07 | 60.00± 0.82 | 65.38± 0.38 | 68.47± 0.15 |
| TALAS | 89.79± 0.03 | 73.87± 0.17 | 61.75± 0.61 | **85.99**± 0.03 | 82.07± 0.09 | 65.79± 0.04 | 75.67± 0.05 | **73.91**± 0.03 | 78.31± 0.04 | 76.35± 0.08 |
| **GATE-KD** | **91.68**± 0.09 | **74.43**± 0.21 | **66.88**± 0.36 | 85.00± 0.06 | **83.46**± 0.02 | 67.44± 0.04 | **77.94**± 0.06 | 72.17± 0.02 | **78.36**± 0.07 | **77.48**± 0.04 |

Table 9 shows that GATE-KD attains the highest mean Avg. in all three teacher–student configurations. It exceeds TALAS by 0.07, 0.27, and 1.13 points in configurations (a), (b), and (c), respectively. The margin is narrow for the two larger students and remains largest for the compact student.

# F Zero-Shot Retrieval

We evaluate the models distilled in the main comparison on ArguAna, SciFact, SciDocs, NFCorpus, and FiQA-2018 using exhaustive cosine ranking. Evaluation uses no instruction prefix or task-specific retrieval training and reports nDCG@10. Retrieval scores are not included in the sentence-level averages of Table 1. Teacher and student-base scores are deterministic, while distilled-student scores are mean ± sample standard deviation over three seeds.

Table 10: Zero-shot retrieval on five BEIR benchmarks (nDCG@10). Distilled-student scores are mean ± sample standard deviation over three seeds; best and second-best results are bold and underlined.

| Method | ArguAna | SciFact | SciDocs | NFCorpus | FiQA-2018 | Avg. |
|---|---|---|---|---|---|---|
| *(a) Qwen3-Embedding-4B → BERT-base (109M)* | | | | | | |
| Teacher | 74.66 | 73.75 | 25.69 | 36.23 | 50.15 | 52.10 |
| Student base | 12.65 | 0.89 | 0.38 | 1.33 | 0.05 | 3.06 |
| RKD | 39.68± 0.25 | 9.74± 0.64 | 1.19± 0.04 | 4.30± 0.30 | 2.82± 0.09 | 11.55± 0.18 |
| Stella | 39.12± 0.17 | 14.29± 0.17 | 3.79± 0.26 | 6.21± 0.14 | 5.57± 0.04 | 13.79± 0.03 |
| CDM | 33.62± 0.27 | 15.11± 0.20 | 3.28± 0.14 | 6.26± 0.09 | 5.08± 0.22 | 12.67± 0.04 |
| DSKD | 33.70± 0.39 | 15.75± 0.49 | 3.35± 0.11 | 6.21± 0.17 | 5.34± 0.24 | 12.87± 0.27 |
| EMO | 21.90± 0.34 | 8.84± 0.28 | 1.87± 0.11 | 4.87± 0.16 | 2.04± 0.06 | 7.91± 0.07 |
| TALAS | **48.95**± 0.17 | **19.94**± 0.09 | **5.70**± 0.12 | **7.94**± 0.08 | **8.15**± 0.02 | **18.14**± 0.07 |
| **GATE-KD** | 45.20± 0.32 | 19.25± 0.66 | 2.81± 0.09 | 6.79± 0.22 | 6.56± 0.21 | 16.12± 0.26 |
| *(b) BGE-M3 → MiniLMv2-H768 (66M)* | | | | | | |
| Teacher | 54.01 | 64.16 | 16.37 | 31.80 | 41.12 | 41.49 |
| Student base | 9.32 | 1.15 | 0.44 | 1.22 | 0.08 | 2.44 |
| RKD | 21.27± 0.23 | 7.84± 0.12 | 1.96± 0.05 | 4.31± 0.14 | 2.26± 0.03 | 7.53± 0.09 |
| Stella | 18.46± 0.37 | 3.11± 0.27 | 1.25± 0.06 | 2.29± 0.15 | 1.66± 0.08 | 5.35± 0.16 |
| CDM | 15.83± 0.19 | 3.77± 0.38 | 0.88± 0.02 | 2.88± 0.16 | 1.41± 0.21 | 4.95± 0.15 |
| DSKD | 16.25± 0.15 | 4.05± 0.10 | 0.88± 0.03 | 2.96± 0.11 | 1.59± 0.07 | 5.15± 0.07 |
| EMO | 13.10± 0.27 | 2.83± 0.17 | 0.72± 0.04 | 2.25± 0.07 | 0.85± 0.09 | 3.95± 0.08 |
| TALAS | **30.14**± 0.32 | 6.55± 0.34 | 1.90± 0.08 | 2.82± 0.18 | 2.83± 0.09 | 8.85± 0.17 |
| **GATE-KD** | 25.30± 0.02 | **11.22**± 0.27 | **3.31**± 0.02 | **4.93**± 0.06 | **4.83**± 0.01 | **9.92**± 0.05 |
| *(c) Qwen3-Embedding-0.6B → MiniLMv2-H384 (22M)* | | | | | | |
| Teacher | 68.11 | 68.26 | 15.72 | 29.73 | 38.68 | 44.10 |
| Student base | 9.85 | 0.66 | 0.32 | 1.28 | 0.05 | 2.43 |
| RKD | 30.12± 0.79 | 5.57± 0.76 | 1.77± 0.12 | 3.34± 0.07 | 1.88± 0.06 | 8.54± 0.33 |
| Stella | 24.49± 0.45 | 2.65± 0.09 | 1.09± 0.07 | 2.62± 0.07 | 1.01± 0.11 | 6.37± 0.09 |
| CDM | 21.09± 0.30 | 1.66± 0.17 | 1.13± 0.03 | 2.57± 0.05 | 0.89± 0.02 | 5.47± 0.06 |
| DSKD | 21.56± 0.38 | 1.77± 0.20 | 1.18± 0.06 | 2.63± 0.03 | 0.89± 0.07 | 5.61± 0.13 |
| EMO | 17.18± 0.14 | 1.41± 0.09 | 0.92± 0.02 | 1.96± 0.06 | 0.70± 0.03 | 4.44± 0.05 |
| TALAS | 35.21± 0.52 | 4.17± 0.09 | 1.44± 0.04 | 3.12± 0.10 | 1.87± 0.07 | 9.16± 0.14 |
| **GATE-KD** | **36.18**± 0.09 | **10.26**± 0.21 | **3.44**± 0.06 | **4.30**± 0.06 | **3.57**± 0.03 | **11.55**± 0.06 |

GATE-KD achieves the highest retrieval average in configurations (b) and (c). In configuration (a), it ranks second to TALAS, with 16.12 versus 18.14 Avg. All distilled students remain substantially below their teachers, indicating that the sentence-level distillation objectives do not fully preserve the teachers' zero-shot retrieval capability.

## G REPRESENTATION DIAGNOSTICS

We examine whether final-embedding supervision changes the internal representations of the student even though GATE-KD applies no intermediate-layer objective. The probe contains 512 unique sentences sampled from the validation splits after excluding sentences found in the distillation corpus. For every teacher and student Transformer layer, we compute centered linear CKA between their pooled representations (Kornblith et al., 2019). Results for the trained students are averaged over seeds 42, 43, and 44. We omit the embedding-output layer because special-token pooling makes this representation constant across examples, for which centered CKA is undefined.

Figure 1 shows that the pretrained students already share substantial representation structure with their teachers. Raw CKA alone therefore combines correspondence inherited from pretraining with changes induced by distillation.

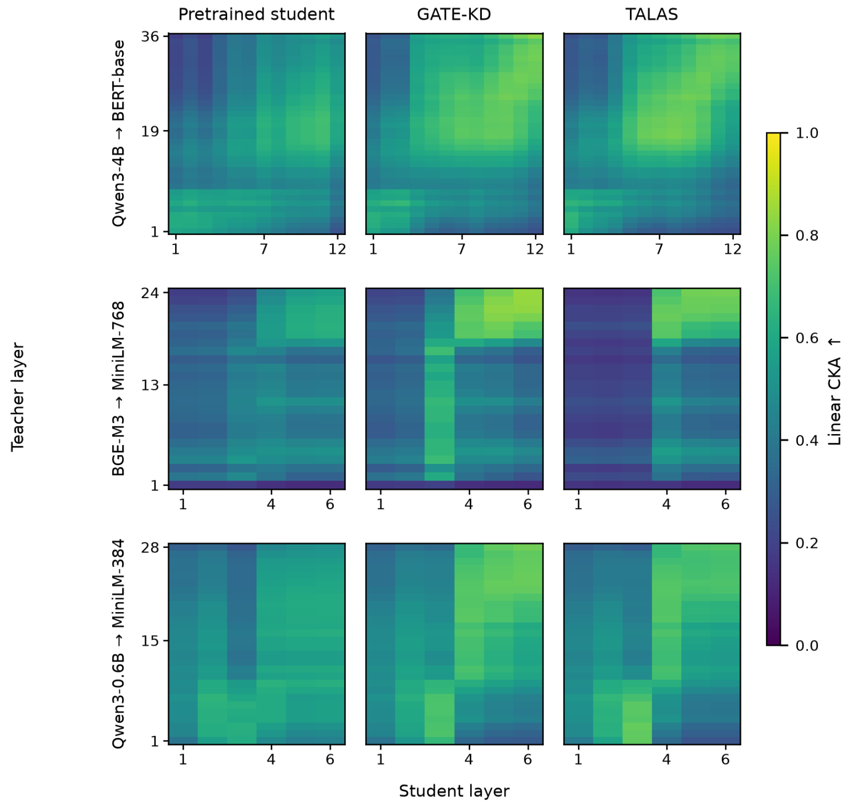


Figure 1: Layerwise linear CKA for the pretrained student, GATE-KD, and TALAS across three configurations. Trained results are averaged over three seeds.

To isolate the training-induced change, Figure 2 subtracts the pretrained-student CKA matrix from the mean matrix after distillation. GATE-KD produces higher final-to-final CKA than TALAS in all three configurations. The raw values are 0.8266, 0.8447, and 0.7845 for GATE-KD, compared with 0.7741, 0.7901, and 0.7333 for TALAS in configurations (a), (b), and (c), respectively. The corresponding changes from initialization are 0.2992, 0.2773, and 0.2781 for GATE-KD, compared with 0.2468, 0.2227, and 0.2269 for TALAS. The final-layer advantage is consequently consistent across the three configurations and ranges from 0.051 to 0.055 CKA.

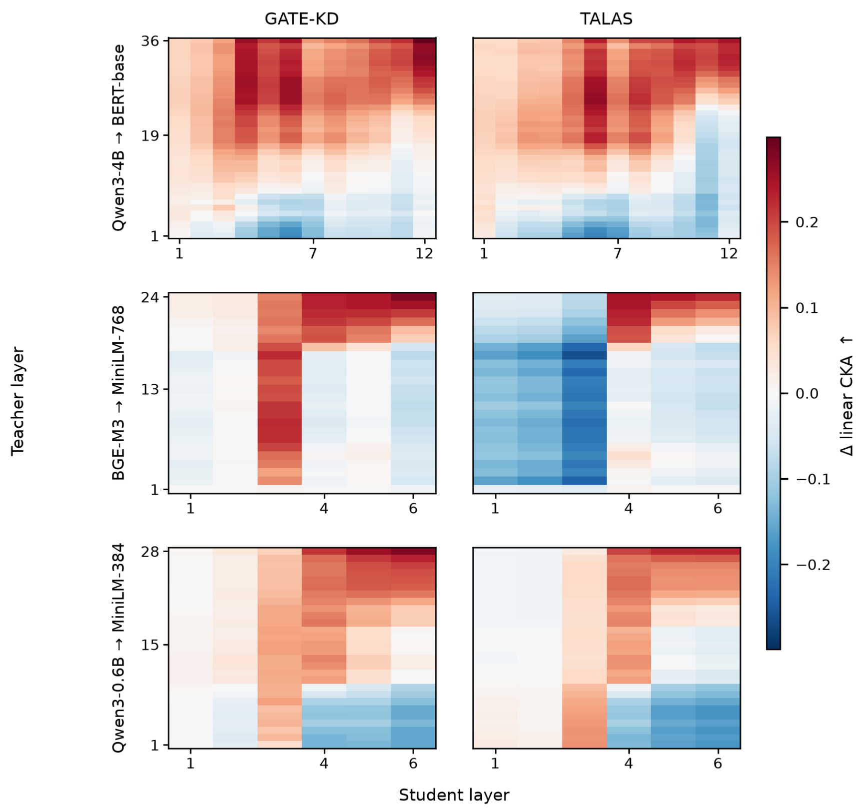


Figure 2: Change in layerwise linear CKA relative to the pretrained student. Red and blue denote increased and decreased similarity, respectively.

The changes are distributed across multiple later student layers rather than being confined to a single teacher–student layer pair. This pattern is consistent with endpoint supervision inducing an internal reorganization of the student despite the absence of explicit intermediate-layer matching. Some layer pairs decrease in CKA, so the result does not imply uniformly stronger alignment at every depth. Moreover, CKA is descriptive and does not establish that the observed internal changes cause the downstream improvements.

# H PCA Centering, Whitening, and Random-Subspace Controls

We isolate preprocessing of the fixed endpoint interface in configuration (c). Every row uses the endpoint-only objective, the same student, corpus, optimizer, training schedule, and epoch-wise Procrustes updates. These are endpoint-only controls, so the reference row is not labeled as GATE-KD or Ours; those names refer to the full endpoint-plus-$H_0$ method in the main tables. The reference estimates its principal directions from the centered cache but applies the resulting linear map without subtracting the cache mean. "Mean-subtracted PCA" applies that mean subtraction as in the textbook affine PCA transform, while "PCA whitening" additionally rescales each retained coordinate by the inverse square root of its fitted eigenvalue. The random controls replace the PCA subspace with three independent Haar-random subspaces of the same rank. Results are averaged over the same three training seeds within each row; random-map draws remain separate rather than being treated as additional training seeds.

Mean subtraction causes the largest degradation, particularly on the six OOD benchmarks. Whitening recovers part of that loss and improves Avg. IOD over the reference, but remains lower on Avg. OOD and overall Avg. The PCA reference exceeds all three random draws despite retaining the same rank. Uncentered SVD attains a numerically indistinguishable overall score, differing from the Table 6 reference by 0.01 points. We therefore retain the evaluated interface, while inter-

Table 11: PCA preprocessing and subspace controls with endpoint-only supervision and Procrustes alignment in configuration (c). Scores are mean $\pm$ sample standard deviation over three seeds; random-subspace draws are shown separately. Best values are bold.

| Interface | Retained energy ↑ | Avg. IOD ↑ | Avg. OOD ↑ | Avg. ↑ |
|---|---|---|---|---|
| PCA + Procrustes (reference) | 0.928 | 67.76$\pm$ 0.05 | 77.54$\pm$ 0.07 | **74.28**$\pm$ 0.03 |
| Mean-subtracted PCA | 0.928 | 66.85$\pm$ 0.15 | 74.95$\pm$ 0.02 | 72.25$\pm$ 0.06 |
| PCA whitening | 0.928 | **68.14**$\pm$ 0.33 | 75.83$\pm$ 0.08 | 73.26$\pm$ 0.16 |
| Random subspace (draw 0) | 0.381 | 67.38$\pm$ 0.07 | 77.28$\pm$ 0.06 | 73.98$\pm$ 0.06 |
| Random subspace (draw 1) | 0.376 | 67.41$\pm$ 0.17 | 77.47$\pm$ 0.05 | 74.12$\pm$ 0.04 |
| Random subspace (draw 2) | 0.374 | 67.48$\pm$ 0.07 | 76.95$\pm$ 0.08 | 73.79$\pm$ 0.03 |
| Uncentered SVD | **0.945** | 67.59$\pm$ 0.22 | **77.61**$\pm$ 0.01 | 74.27$\pm$ 0.08 |

preting this control as evidence for preserving the mean component at application time rather than as evidence that fit-time centering is uniquely optimal.

# I Target-Rank Sensitivity at Fixed Student Width

We isolate the subspace selected by the endpoint interface in configuration (c). The student architecture, 384-dimensional output, distillation corpus, optimizer, training schedule, and Procrustes updates remain fixed, while the retained PCA rank varies over $k \in \{64, 128, 256, 384\}$. This experiment uses the endpoint-only objective. Targets with $k < 384$ are embedded in the native student space by appending zero coordinates before Procrustes alignment. This padding preserves their Gram matrix and active rank, and does not change the capacity or width of the student.

Retained energy is measured on the complete teacher cache. On a fixed sample of 2,048 normalized teacher embeddings $T$, we measure the normalized squared Gram distortion $\|TT^\top - Y_kY_k^\top\|_F^2/\|TT^\top\|_F^2$, where $Y_k$ is the normalized rank-$k$ PCA target. We also report the normalized rank floor $\sum_{j>k}\sigma_j(T)^4/\sum_j\sigma_j(T)^4$ from Equation 7. The Gram quantities are independent of the orthogonal gauge and are therefore computed once, while downstream scores are averaged over three seeds.

Figure 3 shows monotonic changes in all three quantities. Increasing $k$ from 64 to 384 raises retained teacher energy from 59.32% to 92.84%, reduces normalized Gram distortion from 6.797% to 0.086%, and improves Avg. from 72.57 to 74.27. The intermediate ranks 128 and 256 attain 73.77 and 74.21 Avg., respectively. Most of the downstream benefit is therefore recovered by rank 256, with the final increase to rank 384 contributing only 0.06 points in this configuration.

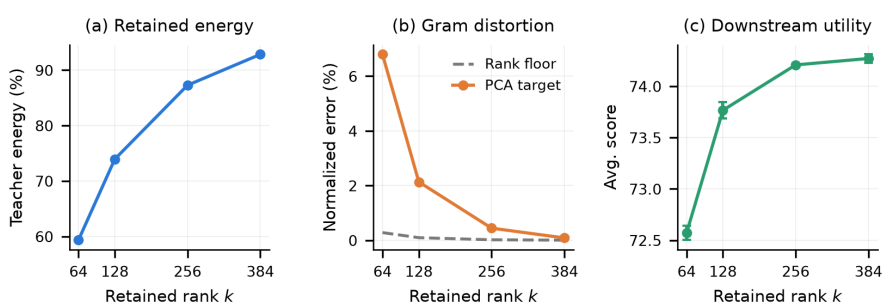


Figure 3: Sensitivity to retained PCA rank at fixed student width in configuration (c). Downstream scores show mean $\pm$ sample standard deviation over three seeds.

PCA distortion remains roughly 21–25 times above the rank floor because its objective differs from normalized Gram approximation; the floor bounds irreducible error rather than predicting this interface's distortion. Increasing rank improves geometry and utility with diminishing returns. With the student architecture fixed, this is target-rank sensitivity, not a width ablation.